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Detection of nocturnal epileptic seizures using a wearable armband: A deep learning approach combining accelerometry and photoplethysmography signals

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ABSTRACT

Background: Epileptic seizures can lead to severe outcomes including sudden unexpected death in epilepsy (SUDEP). Clinical standard for seizure diagnosis and detection requires electroencephalography and video monitoring, which is yet considered not suitable for home use, especially during nighttime sleep in a low-light condition. We proposed a deep learning (DL)-based approach to automatically detect nocturnal major seizures using a wearable armband that can potentially help reduce SUDEP risk through timely caregiver intervention.
Methods: In this prospective cohort study, 68 patients with major seizures were monitored for up to three months using a wearable armband (NightWatch®) capturing tri-axial accelerometry (ACM) and photoplethysmography (PPG) signals. A two-step approach was designed: (1) a pre-screening step using threshold-based algorithms to identify suspected seizure events (ACM standard deviation >0.4 or heart rate increase >10%), and (2) a DL model (CNN-LSTM with attention mechanism) to recognize true seizures. Model performance was evaluated via a 10-fold cross-validation, reporting sensitivity (SEN), false alarm rate (FAR), and area under the ROC curve (AUC).
Results: In 788 overnight recordings (6304 hours), a total of 1846 severe seizures were identified. The pre-screening step achieved 0.940 sensitivity in pre-identifying or 'preserving' seizures, reducing data volume by 81% (from 6304 to 1201 hours). The DL model demonstrated a mean accuracy of 0.793 [95% CI: 0.745–0.841], a mean sensitivity of 0.762 [95% CI: 0.704–0.821], a mean positive predictive value of 0.334 [95% CI: 0.229–0.356] and a mean false alarm rate of 0.165/hour [95% CI: 0.097–0.234]. These results exceeded those of single (signal) modality detection methods.
Conclusion: Our two-step approach enables accurate, long-term detection of severe nocturnal seizures in home settings. The wearable system provides a practical solution for continuous monitoring and real-time alerts, thus potentially reducing SUDEP risk and improving patient safety, fulfilling an urgent unmet need in epilepsy care. Furthermore, by enabling long-term home monitoring, this system may help assess the relationship between seizure events and lifestyle-related triggers such as sleep deprivation, stress, physical exertion, or alcohol consumption, thereby supporting the development of personalized preventive strategies.

1. Introduction

Epilepsy is a chronic neurological disorder characterized by abnormal, excessive neuronal discharges, leading to recurrent seizures, requiring patients to do regular review and long-term treatment. Epilepsy has a high incidence rate and a wide range of affected people, bothering patients of different ages, races, and classes around the world. A systematic review and meta-analysis of shows that the global

Abbreviations: ACM, accelerometry; AUC, area under the ROC curve; CNN, convolutional neural network; ECG, electrocardiography; EEG, electroencephalography; FAR, false alarm rate; HR, heart rate; LSTM, long short-term memory; PPG, photoplethysmography; ROC, receiver operating characteristic; SD, standard deviation; SEN, sensitivity; SUDEP, sudden unexpected death in epilepsy; SVM, support vector machines.

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prevalence of epilepsy ranges from 5.8 % to 7.3 % [1]. It is estimated that there are currently about 50 million active epilepsy patients in the world (referring to patients with frequent epileptic seizures needing to maintain anti-epileptic treatment), and among them 30 % are refractory [2]. Epileptic seizures have many potential dangers, including injury, status epilepticus, and sudden unexpected death in epilepsy (SUDEP) [3]. The sudden death rate in epilepsy patients is 20 times higher than that in the general population [4]. Among people with refractory epilepsy, the incidence rate of SUDEP is 1.1–5.9 per 1000 people per year, and the frequency of epilepsy patients with indications for surgery or epilepsy patients who still have seizures after surgery is 6.3–9.3 per 1000 people per year [5]. They will face the risk of epilepsy attacks at any time for a long time or even throughout their lives. Therefore, long-term seizure detection methods that can be used out-of-hospital would greatly improve the ability of people with epilepsy to live independently.

Studies have shown that before and during an epileptic seizure onset, there are several characteristics significantly different from other stages, such as headache, chest tightness, sleep disorders, and irritability [5]. These symptoms are often reflected in multiple physiological signals, and timely identification is crucial for detecting seizures. Pre-ictal symptoms may even be able to predict the onset of epilepsy. In 2017, the American Academy of Neurology and the American Epilepsy Society jointly issued a practice guideline about the incidence and risk factors of SUDEP, which pointed out that if monitoring equipment is used, the caregivers of the patients can be notified promptly when an epileptic seizure occurs [6]. If appropriate measures are taken, the chances of the occurrence of respiratory dysfunction and hypoxemia will be reduced.

SUDEP usually occurs within a short time of a convulsive seizure (2–17 minutes, average 10 minutes) and may result in a hypo ventilatory state or asphyxia, accompanied by normal arousal dysfunction [5]. Because of the uncertainty of the time and place of epileptic seizures, patients may have seizures at any time and harm themselves. Therefore, caregivers should provide constant supervision to ensure the safety of the patient. Especially at night, it is difficult for caregivers to be with patients all the time. In this context, wearable devices provide a more convenient option for both caregivers and patients, allowing for greater independence and safety in patients' daily lives.

At present, the work of detecting epileptic seizures mainly focuses on identifying abnormal changes in patients' electrocardiography (EEG) [7–10], clonus [11,12], tachycardia [13,14], and other abnormal symptoms in respiration rate and sounds [15]. Compared with short-term episodic detection methods such as video-EEG used in clinical practice, continuous detection is an approach for prolonged monitoring of epileptic patients, providing timely detection of seizures.

In clinical settings, video-EEG systems are employed for patient monitoring [12], which is the gold standard for epileptic seizure monitoring. However, the placement of EEG devices on patients' scalps may induce discomfort and allergic reactions, while restricting their activities. Consequently, video-EEG monitoring is not suitable for long-term daily use. Considering that continuous monitoring of epilepsy patients must not affect patients' normal life, using wearable sensors is considered promising because of its good portability and non-invasiveness.

Body movement is considered a preferred choice for analysis among all non-EEG modalities because clear and distinguishable features can be observed in movement signals during epileptic seizures characterized by the associated motor phenomena [16]. In motor seizures, the distinctive movements are often composed of patterns similar to those induced by electrical stimulation of the primary motor areas of the brain. Motor seizures typically consist of a series of single or multiple movement patterns, namely myoclonic jerks, clonic movements, and tonic movements [17]. Moreover, the most straightforward manifestation of movement is reflected in the changes in the accelerometry (ACM) signal [18,19]. In addition to body movement, the ictal tachycardia is also an important physiological characteristic of convulsive seizures (including bilateral tonic-clonic seizures and generalized tonic-clonic seizures) which is particularly common in temporal lobe epilepsy. The incidence of ictal tachycardia is about one-third of epileptic seizures [20] and may even occur a few seconds before body movements. Abnormal changes in HR can be used for early seizure detection. Studies employing electrocardiography (ECG) for seizure detection have predominantly concentrated on patients in hospital settings [21–23]. The data analysis primarily focuses on the magnitude and duration of HR elevation, often involving the extraction of time- and frequency-domain features within each heartbeat cycle or a fixed-length sliding window. Jeppesen et al. [23] used wearable sensors to acquire ECG signals for seizure detection. They extracted a total of 20 features such as second-order central difference of HR and cardiac sympathetic index, achieved a sensitivity (SEN) of 0.8 in detecting seizures using a thresholding classifier. However, since some seizures are not associated with HR changes but with solely body movement, these seizures would be missed when applying an ECG- or HR-based model. Cooman et al. [24] designed a transfer learning algorithm using support vector machines (SVM) and ECG signals, and they found that the false detection rate was 1.9 times per hour and suggested that when combining ECG with other signal modalities such as ACM, the false detection rate could decrease with a factor of 5–10 compared to an ECG-only model. Vandecasteele et al. [22] monitored patients' seizures in a hospital environment using a bipolar ECG device. They achieved promising results in seizure detection by employing an SVM model based on features including the peak, mean, and standard deviation of HR, although the results were slightly worse than the use of ECG. Moreover, Mohammadpour et al. [25] and El Atrache et al. [26] found that the frequency, amplitude, duration, slope, smoothness, and area under the curve of photoplethysmography (PPG) signals would all changed during seizures. This suggested the potential value of using more PPG characteristics rather than only HR for improving seizure detection. However, only using ECG or PPG signals to detect seizures seem not sufficient. In the case of epileptic seizures, HR related measurements may be combined with body motion related measurements to ensure greater accuracy [27].

Traditional statistical analyzing methods rely on manual feature extraction, which demands significant human efforts. Moreover, handcrafted feature extraction may not encapsulate all the pertinent information necessary for effective seizure detection. DL methods offer the advantage of automatically processing data by learning and extracting features directly from signals. Moreover, they could discern potential relationships and interactions among different dimensions within the data. This capability enables DL models to capture complex patterns and dependencies that may not be readily apparent through manual feature extraction or traditional analysis techniques. In this study, we adopt a hybrid design: raw ACM and PPG signals are first transformed into a set of statistical and physiological features, which are then processed by a DL model. This approach combines the interpretability and computational efficiency of feature engineering with the pattern recognition capabilities of DL, potentially improving seizure detection performance by incorporating additional PPG characteristics alongside HR. Similar feature engineering and integration strategies have also demonstrated clinical value in other disease prediction domains [28].

Although various seizure detection methods have been proposed using EEG, ECG, ACM signals, and others, most of them are limited to in-hospital settings or rely on single-modality inputs. Most previous studies were also based on short-term datasets collected over a few hours or days, limiting their ability to reflect long-term usability and reliability. Furthermore, few studies focus on continuous monitoring during nocturnal sleep for a long period in real-world, out-of-hospital environments. Existing methods either lack generalizability, impose discomfort due to sensor placement (e.g., EEG), or demonstrate high false alarm rates when deployed in wearable devices. These limitations highlight a significant gap in developing seizure detection systems that are not only accurate and multimodal, but also suitable for real-world, long-term use.

In contrast, our work leverages a long-term dataset spanning several months, providing real-world evidence for system performance and robustness under daily-life conditions. To address these challenges, this study aims to develop a two-stage seizure detection framework using multichannel or multimodal signals (ACM and PPG) collected from a wearable armband. The proposed system includes a lightweight pre-screening module to reduce computational load and a CNN-LSTM-based classifier with attention mechanism for accurate seizure recognition. Our objective is to enable robust, low false-alarm, and real-time detection of severe nocturnal seizures in out-of-hospital settings, thereby supporting independent living for people with epilepsy.

Comparing with some related recent studies as shown in Table 1, the main contributions of this study are summarized as follows:

- Multimodal Signal Fusion: We integrate tri-axial ACM and PPG to capture both movement and cardiovascular changes during seizures, enabling richer feature representation.
- Two-Stage Detection Framework: A lightweight pre-screening algorithm filters out irrelevant segments with 94 % sensitivity while reducing data volume by 81 %, followed by a CNN-LSTM model with attention to classify seizure events.
- Improved Detection Performance: Our system achieves a sensitivity of 76.2 %, a false alarm rate of 0.165/hour, and an AUC of 0.793 on a real-world dataset spanning 6304 hours from 68 patients, outperforming traditional single-modality methods.
- Scalable Wearable Application: The proposed approach supports continuous, unobtrusive seizure monitoring in home environments, demonstrating the feasibility of deploying DL models in real-time healthcare settings.

The paper is organized as follows. Section 2 describes the dataset and data acquisition protocol. Section 3 presents the signal preprocessing techniques, the pre-screening strategy, and the architecture of the DL model. Section 4 provides experimental results and comparative analysis. Discussions, conclusions, limitations, and future work are given in Section 5 and 6.

2. Data collection and annotation

In this prospective study, nocturnal tri-axial ACM and PPG recordings were continuously collected using a NightWatch® armband (LivAssured, Leiden, The Netherlands) from 68 patients during the night. Each patient was monitored for two to three months in the Kempenhaeghe Academic Center for Epileptology, the Netherlands. As shown in Fig. 1, signals were collected with the armband and transmitted to a base station. The NightWatch system proved a sensitive and convenient monitoring method for epileptics with convulsive seizures, especially for those who had more seizures at night while releasing their caregivers from accompanying the patients all day long.

Table 1
Related recent studies in seizure detection highlighting signal modalities, clinical scenarios, and model types.

Study (Year)	Signal Modality	Scenario	Seizure Type	Model Type
Ahmad et al. (2024) [7]	EEG (publicly accessible database)	In-hospital	Not mentioned	DL classifiers
Qiu et al. (2023) [8]	EEG (publicly accessible database)	In-hospital	Not mentioned	DL classifiers
Yang et al. (2021) [11]	Video	In-hospital	GTCS	DL classifiers
Johansson et al. (2019) [12]	3-axis ACM	In-hospital	TC	Single ML classifier
Vandecasteele et al. (2017) [22]	ECG and PPG	In-hospital	focal seizure	Single ML classifier
Touserkani et al. (2020) [25]	PPG	In-hospital	GTCS	Statistical analysis
Our proposed method	ACM and PPG	Out-of-hospital	Severe seizures	DL classifiers

Note: DL = deep learning; ML = machine learning; TC = tonic-clonic; GTCS = Generalized tonic-clonic seizure.

The collected raw ACM and PPG signals had a sampling rate of 11–12 Hz and 100 Hz, respectively. To ensure signal quality and consistency, any nightly recording containing sampling dropouts was excluded from analysis. This quality control process resulted in a final dataset comprising 788 validated nights, totaling approximately 6304 hours of recordings. Both ACM and PPG signals were subsequently resampled to 20 Hz to unify the temporal resolution for downstream processing.

Seizure annotations were made by experienced clinicians and trained nurses based on synchronized video and audio recordings. The annotated events were categorized into three clinically significantly severe types: (1) tonic-clonic seizures, (2) major hypermotor seizures, and (3) prolonged tonic seizures exceeding 30 seconds. Minor seizures, including myoclonic or short tonic events, were not included as alarm targets in this study. To ensure labeling consistency, 10 % of the annotations were randomly selected and cross-reviewed by clinicians for quality assurance. The final dataset was structured for supervised learning, with seizure and non-seizure segments labeled accordingly.

3. Methods

The proposed seizure detection framework consists of two steps. Fig. 2 presents the algorithmic structure and technical pipeline of the proposed two-stage seizure detection framework, including (A) the pre-screening module, (B) the feature extraction process, and (C) the deep learning classifier architecture. The first step is a lightweight pre-screening step aiming at identifying suspected seizure events as many as possible with a simple thresholding algorithm, where a large part of recordings corresponding to non-seizure periods can be excluded for further analysis. This can prevent feeding all data into deep neural networks for model learning which is expected to be of high computational load. Instead, a subset of data after pre-screening remains to be used for DL. Importantly, the pre-screening should be, on purpose, allowed to miss no or only a small number of actual seizure events. Among the suspected severe seizures, there could be severe seizures, minor seizures, and other non-seizure activities such as turning over in the bed, sitting up or waking up. The second step is the seizure detection step, during which a DL model is trained on the data of the suspected events to detect severe seizure events from all other suspected events. In this paper, for seizure detection, seizures are referred to severe seizures excluding minor seizures, and non-seizure events are referred to the remaining suspected events including non-seizures and minor seizures.

Fig. 1. The NightWatch wearable armband and base station for nocturnal seizure monitoring.

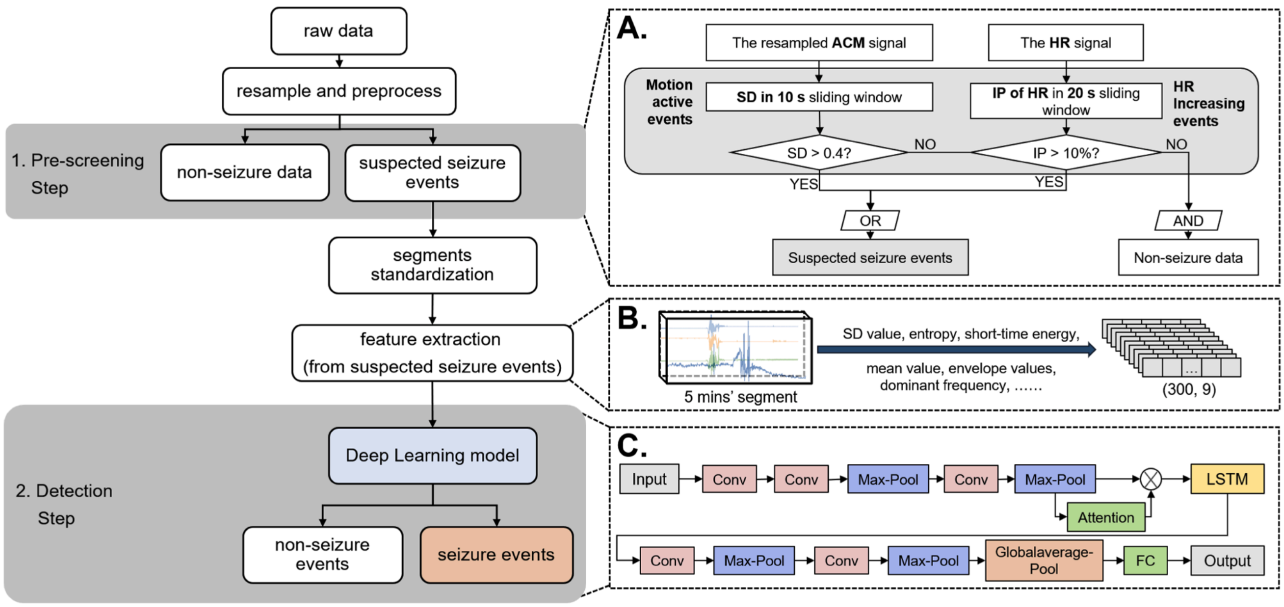

Fig. 2. Technical architecture of the proposed two-stage seizure detection framework.
(A) Pre-screening module for extracting suspected seizure segments based on accelerometry and HR signals. (B) Feature extraction process for standardized 5-minute signal segments. (C) DL model architecture combining CNN, LSTM, and attention mechanism for final classification.

3.1. Pre-screening of suspected severe seizure events

To reduce computational overhead and focus model training on seizure-relevant segments, we implemented a rule-based pre-screening strategy that extracts potential seizure episodes based on changes in motion and HR signals. For PPG signals, a second order Butterworth bandpass filter (0.5–3.5 Hz) was applied to reject noise and frequency components outside the possible HR range. The choice of cutoff frequencies was to ensure the pass band corresponding to the human HR range from 30 to 210 beats per minute (BPM). For ACM signals, an eighth-order Butterworth lowpass filter with a 10 Hz cutoff was applied to remove high frequency interference. HR was estimated from the PPG signal by detecting the systolic peaks. Both ACM and HR signals were then resampled to 20 Hz to ensure temporal alignment.

In our previous study [29], we found that the standard deviation (SD) of the ACM signal that conveys motion artifacts is crucial in assessing a subject's body movement. In addition, we found that some TC seizures are associated with increased HR, which can sometimes occur even before the body movements. Therefore, in the pre-screening step, to identify suspected events, we delved into quantifying both HR and motion changes.

In this study, we employed a dual-threshold strategy to identify 'movement active' and 'HR increased' segments from the resampled ACM and HR signals. As shown in Fig. 2. A, a sliding window of 10 s was used to calculate the SD of ACM signals, and a sliding window of 20 s was used to calculate the HR's increasing percentage (IP), compared with the HR in the last sliding window, as shown in the following equation:

$$IP\ (\%) = \frac{\mathrm{HR}(i) - \mathrm{median\_HR}}{\mathrm{median\_HR}}, \quad (1)$$

where the HR(i) is the HR value at the time i [ranges from 0 to 1200 (in total 20 minutes)], and *median_HR* is the median value of the HR during the whole recording segment. If the SD is higher than 0.4 or if the IP is over 10 %, we considered that there was a suspected severe seizure event, and we combined the 'motion active' events and the 'HR increased' event together, as shown in Fig. 3. The dashed lines in the upper and lower graphs represent segments with 'movement active' and segments with 'HR increased', respectively. If there is partial overlap

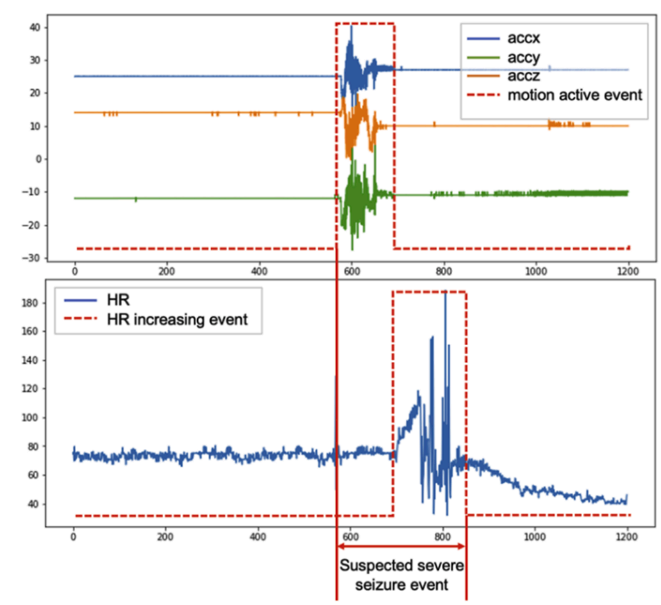

Fig. 3. Example of identifying suspected seizure segments based on ACM and HR signals. 'accx, accy, accz' are the ACM signal on x, y and z axis.

between the two types of segments, they will be merged into a complete segment, indicating a suspected epileptic seizure segment. Otherwise, they will constitute separate suspected epileptic seizure segments. Therefore, the segments shown in this figure represent a segment that includes both 'movement active' and 'HR increased'.

3.2. Seizure labeling and feature extraction

Each suspected severe seizure events obtained from the pre-screening step was assigned a binary label based on clinician-verified annotations. A segment was labeled as a 'suspected severe seizure' if a documented annotations occurred within or near its time window (±2 minutes). Otherwise, the segment was considered a 'non-seizure' event.

To maintain event-level consistency and avoid a real seizure event was divided into too many segments, when the time interval between two seizure segments was less than 100 s, we merged these two segments. When multiple seizure events occurred within a four-minute window around an annotation, we also labeled them all as one seizure event.

To prepare the input for model training, each suspected severe seizure event was transformed into a fixed-length feature representation. Before training the detection model, we first standardized the length of suspected events because most suspected events were in different length. As shown in Fig. 4, most suspected events were less than 5 minutes, so all segments were standardized to a duration of 300 s (5 minutes). Segments shorter than this length were zero-padded, while longer segments were cut. This fixed-length standard ensures consistent temporal input across samples.

From each segment, we extracted nine statistical and frequency-domain features at 1 Hz resolution from the ACM and HR signals, including the SD value, entropy, short-time energy, mean value, the lowest and highest envelope values, and the dominant frequency of the total acceleration, as well as the mean value and SD value of HR. The resulting data structure was a two-dimensional feature matrix of shape (300, 9), capturing the temporal evolution of physiological patterns within each segment. This matrix served as the input to the DL classifier described in the next section.

3.3. Deep learning for seizure detection

To further classify real seizure and non-seizure events within the suspected severe seizure segments, we designed a hybrid DL model that combines convolutional and recurrent layers with an attention mechanism. This architecture leverages both spatial and temporal dependencies within the extracted feature sequences.

As shown in Fig. 2.B and Fig. 2.C, the input to the DL model is a 2D feature matrix of shape (300, 9), representing a 5-minute segment with 1-second resolution and nine features per timestep. The input feature matrix of shape (300, 9) passes through two initial convolutional layers with 16 and 32 filters, respectively, followed by max-pooling to reduce the temporal dimension. A third convolutional layer with 64 filters further extracts hierarchical features, and max pooling is applied again. An attention mechanism is then introduced by generating dynamic weights through a dense layer, which are element-wise multiplied with the feature maps to emphasize significant channels [30]. The resulting attended feature sequence is fed into an LSTM layer with 32 units to capture temporal dependencies. Subsequently, three additional convolutional and max-pooling blocks refine the features. Global average pooling aggregates the feature maps into a fixed-length vector, which is passed through a fully connected layer to produce the final seizure probability prediction.

To prevent overfitting, dropout was applied to the LSTM and dense layers during training. The proposed model was implemented using TensorFlow and trained using the Adam optimizer, with an initial learning rate of 0.001, determined through a preliminary search among values 0.01 to 0.0001, and a batch size of 64. Early stopping with a patience of 10 epochs was employed to prevent overfitting based on validation loss monitoring. The binary cross-entropy loss function was used to optimize the model parameters. To decrease the inherent imbalance between seizure and non-seizure segments in the dataset, we applied class weighting during training, assigning a higher weight to seizure samples in the binary cross-entropy loss function. The weights were determined based on the inverse frequency of each class in the training set, thereby ensuring that seizure events contributed proportionally more to the optimization process. The model was compiled using TensorFlow, Keras, and Python 3.11. The detailed parameters of the DL model and experimental setup are summarized in Table 2.

3.4. Model training and evaluation

Due to significant variations in the number of seizures observed among patients, a subject-independent stratified split strategy for 10-fold cross validation was used to ensure as much as possible a similar number of seizures per fold. Firstly, we sorted the patients in descending order based on their seizure occurrences (i.e., number of seizure events) and separated them into seven groups, each group contained nine or ten patients. Then one patient from each group was taken to constitute a fold of data. At last, we got ten folds of data, and each fold contained six to seven patients.

As shown in Fig. 5, in each training iteration, we selected one-fold to form a testing dataset comprising data from six or seven patients. The remaining folds were then utilized for both training and validation. This stratified split strategy allowed the model to be trained as a patient independent classifier by using one patient's data only for training or only for testing. The amount and some statistics of the seizure events and the

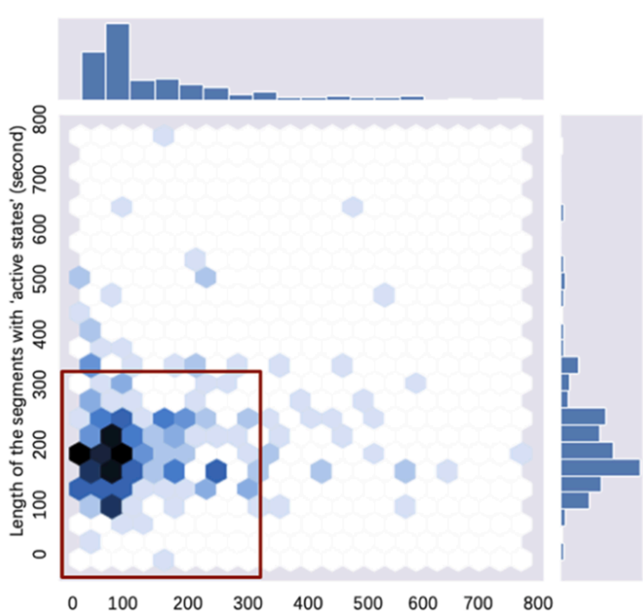

Fig. 4. Segment length distribution for movement-active and heart-rate-increase events.

Table 2
Experimental setup and DL model parameters.

Component	Parameter	Value/Description
Input features	Input shape	(300, 9), 5-min segment at 1 s resolution
	Feature types	SD, entropy, short-time energy, dominant frequency, etc.
DL model architecture	Conv layer 1–3 + Max-pooling	filters: 16, 32, 64; kernel: 2, 20, 4
	Dropout	0.3
	Attention module	Dynamic weights via dense layer × feature maps
	LSTM	32 hidden units
	Conv layer 4–6 + Max-pooling	32→16 filters, kernel=4
	Global Average Pooling	
	Fully connected layer	'sigmoid' activation
Training settings	Optimizer	Adam
	Learning rate	0.001
	Batch size	64
	Epochs	Earl stopping, patience = 10
	Loss function	Binary cross-entropy
Output	Task	Binary classification (seizure vs. non-seizure)

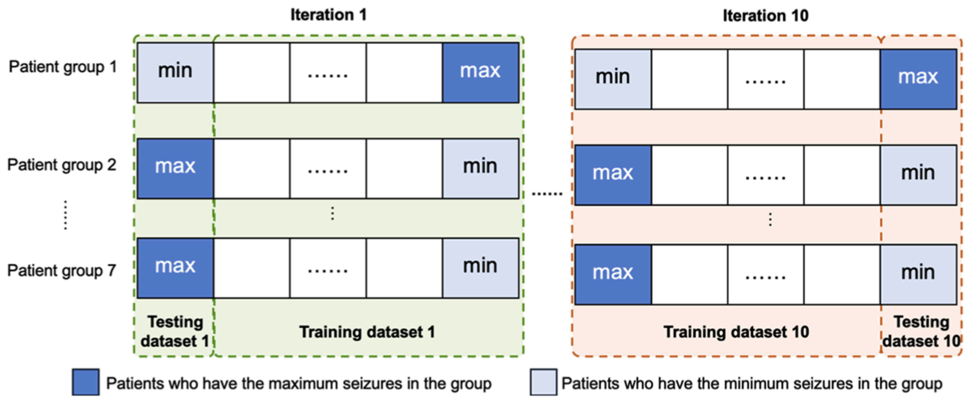

Fig. 5. Subject-independent stratified split strategy in the model training and evaluation step.

monitoring duration of each fold are shown in Table 3.

The results of the detection model in test data were evaluated using receiver operating characteristic (ROC) curves, SEN and false alarm rate (FAR). The ROC curve is a graphical representation used to evaluate the performance of a classifier, which shows the relationship between the true positive rate and the false positive rate at different thresholds. The ROC curve captures the performance of the classifier over the entire prediction range, independent of a particular threshold. The area under the ROC curve (AUC) is a commonly used metric that indicates the magnitude of the classifier's ability to discriminate between all positive and negative samples. In this work, seizure events were considered positive samples while non-seizures events were negative samples.

4. Results

In this study, we collected 1963 annotations from 68 patients during 788 nights and recorded the onset time annotation of each seizure event. All the annotations were determined by the clinicians according to abnormal body movements, HR raises, sounds, and observations. We then obtained seizure events labeled according to the annotations combined with the proposed pre-screening method. The average amount of seizure events per patients was 27.14 ± 63.55 (range from 1 to 372). The average monitoring time was 11.79 ± 19.25 nights (range from 1 to 115).

By using the proposed pre-screening method, a total of 14,417 suspected severe seizure events from 68 patients were identified, where 1109 (7.69 %) of which were TC seizure events, 564 (3.91 %) were major seizure events and 173 (1.20 %) were other tonic seizures. The SEN of identifying seizure events during the pre-screening step was 0.940 and the PPV was 0.128. To be more specific, in the pre-screening step, a total of 1846 severe seizure events were successfully identified, and 117 annotations were not presented.

To illustrate the time difference between HR increases and body movements, we plotted the percentage of increase in HR and the starting time of body movement in Fig. 6. In this figure, the dark blue line showed the mean HR IP, and the light blue shadow area showed the standard deviation. The red line corresponded to the movement starts of seizures recorded by the NightWatch. The red dashed line indicated that, on average, the HR increase already started about 100 s earlier than seizure-related body movement. Interestingly, the moment when the HR increase reached its peak was close to the recorded start of movements.

In the seizure detection step, by using stratified split strategy and 10-fold cross validation to train and test the proposed attention-based CNN-LSTM algorithm, we obtained the ROC curve for each test fold, as shown in Fig. 7. It can be seen that the mean AUC over folds was 0.793 [95 % CI: 0.745–0.841], ranging from 0.647 to 0.902. To further examine the contribution of each component, we conducted a simplified ablation study comparing single-modality inputs and the full proposed framework, with the results summarized in Table 4.

Furthermore, we calculated SEN and PPV for each fold of patients, and the results was presented in Fig. 8. The SEN and PPV varied among patients, so did the number of seizures that the patient had. The size of the shadow around the spot indicated the number of the sever seizure events that patient had. The best performance showed the SEN and PPV were both 1.0, and the worst performance showed the SEN and PPV were both 0.0. The achieved mean SEN was 0.762 [95 % CI: 0.704–0.821], and the mean PPV was 0.334 [95 % CI: 0.229–0.356]. In addition, the mean FAR per hour was 0.165 [95 % CI: 0.097–0.234]. Considering the SEN of 0.940 in the pre-screening step, the overall SEN of our proposed method in detecting severe seizure events for the entire dataset was 0.716 [95 % CI: 0.713–0.772].

Table 5 presented the results of the proposed method in this study in comparison with other HR related sensors for seizure detection [24]. Notably, the SEN of single HR-based sensors, such as ECG and PPG, for seizure detection did not exceed 0.710. Specifically, when relying only on one wearable PPG sensor device, the SEN dropped to 0.320. However, this comparison is intended to provide a qualitative overview of the sensing approaches and application settings, rather than a direct quantitative performance benchmarking.

Table 3
Distribution of seizure counts and monitoring duration across 10-fold partitions.

Fold	Number of patients	Number of seizures	Average number of seizures per patient	Monitoring duration (day)
1	7	95	13.57	52
2	7	95	13.57	75
3	7	103	14.71	70
4	7	115	16.43	72
5	7	123	17.57	55
6	7	132	18.85	61
7	7	160	22.86	75
8	7	291	41.57	138
9	6	315	45.00	95
10	6	409	68.17	95

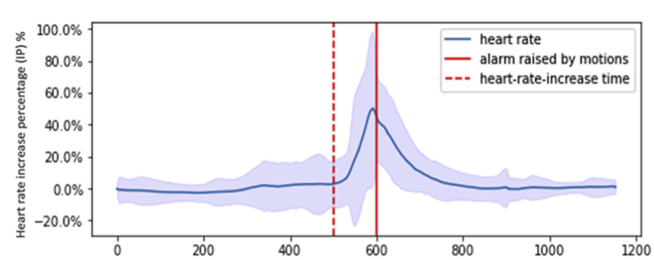

Fig. 6. Temporal alignment of HR changes and movement onset during seizures.

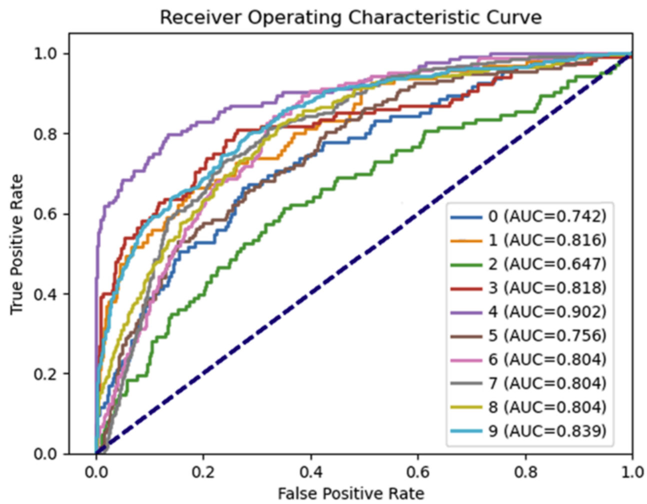

Fig. 7. Receiver operating characteristic (ROC) curves from 10-fold cross-validation.

Table 4
Ablation results show the contribution of each component in the proposed framework.

Research	AUC [95 % CI]	SEN [95 % CI]	PPV [95 % CI]
ACM-only	0.731 [0.592–0.870]	0.634 [0.510–0.757]	0.236 [0.169–0.302]
PPG-only	0.595 [0.481–0.708]	0.596 [0.472–0.721]	0.160 [0.113–0.206]
Proposed method (ACM+PPG)	0.793 [0.745–0.841]	0.716 [0.713–0.772]	0.334 [0.229–0.356]

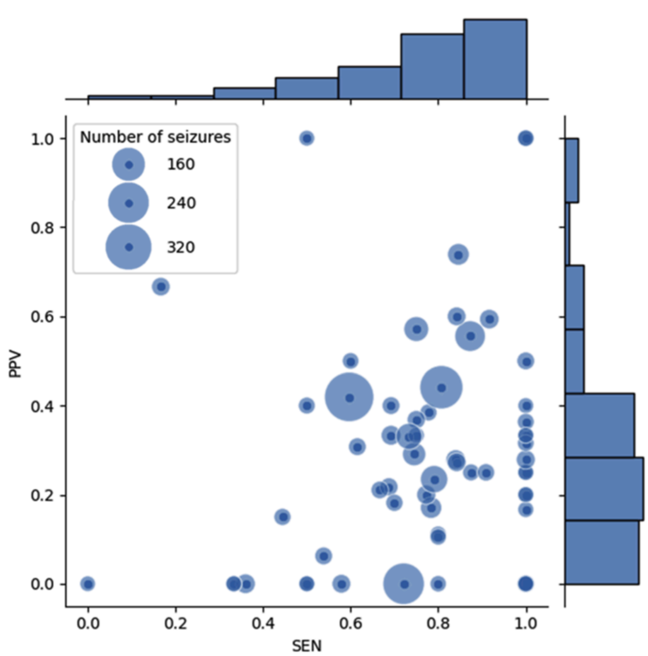

Fig. 8. Sensitivity (SEN) and positive predictive value (PPV) distribution across patients. Dot size represents the number of seizure events per patient.

5. Discussion

In this study, we developed a two-step approach for detecting severe nocturnal epileptic seizures using a combination of ACM and PPG data from an armband, which showed high SEN in the pre-screening step and notable performance in the detection step, and this ensured the reliability of using it in the real scenario. The two-stage strategy was designed to reduce false positives by pre-screening suspected seizure segments before applying the DL classifier. The multimodal classifier design was chosen to leverage complementary physiological cues from accelerometry and photoplethysmography, aiming to capture both motor and cardiovascular changes during seizures.

The proposed two-stage seizure detection framework is intentionally designed to balance modeling complexity, interpretability, and deployment feasibility. The first pre-screening module employs simple statistical methods to rapidly exclude non-informative signal segments, reducing data volume by 81 % while preserving over 94 % of seizure events. The pre-screening module is lightweight and suitable for real-time filtering on wearable devices, with an estimated computational complexity of O(N), where $N = T \times D$ (time steps $\times$ signal dimensions). This indicates that the vast majority of clinically relevant seizures were retained for second-stage classification, and the computational cost was substantially alleviated. These results suggest that the pre-screening step introduces only a minimal trade-off in sensitivity while offering significant gains in efficiency and feasibility for real-time wearable deployment.

The high SEN of pre-screening is critical in clinical applications where minimizing missed seizure events was paramount. However, the reasons for missing were varied. Importantly, the few cases marked as “missed” during pre-screening were primarily due to annotation discrepancies, such as duplicate annotations of different phases of a single prolonged seizure or delayed manual records relative to the true onset, rather than the inability of the algorithm to capture seizure-related physiological changes. In the pre-screening step, we found that one seizure event could be originally annotated multiple times. These multiple annotations were set considering separate parts of a single long seizure event as several separate seizures by the nurses. Also, the annotations were acquired through different methods including the nurses’ diaries and the alarm of clinical devices according to increasing HR and abnormal movements, which also caused duplicated annotation. In other cases, the missed annotations recorded by clinicians were later than the actual time with body movements and/or HR changes. As a result, the gap between the event and the annotation might exceed the threshold we have set. Our pre-screening algorithm treated these duplicated and late annotations as ‘missed’ events. Moreover, in the pre-screening step, we only considered severe seizure annotations, while some minor seizures that do not require alarms were present as well, and those minor seizures might be screened as suspected severe seizures. This approach aimed to highlight the detection of clinically severe seizures, thereby reducing the effects of alarms from minor events.

The second stage uses a hybrid CNN-LSTM network with attention, integrating convolutional layers for local pattern extraction. An LSTM layer for modeling long-range dependencies, and an attention mechanism for dynamic weighting of time points. While this introduces moderate structural complexity, it enables the model to capture temporal progression across HR and movement channels, which is essential to reveal our novel physiological observation that ictal HR changes often precede motor manifestations. In terms of computational complexity, the CNN and attention components are both O(N), while the LSTM introduces a higher cost of $O(N^2)$ due to its recurrent operations over time steps. However, the system remains feasible for deployment considering the data volume reduction from the first step and the moderate size of the input sequence. Before applying DL, we reduce the size of the dataset substantially (from 6304 hours to 1201 hours), which greatly cuts down the computational cost and makes our approach more suitable for real-time applications. The optimal average accuracy of the CNN DL model integrating the attention and LSTM layers was 0.794. This result indicates that our model reliably distinguishes between seizure and non-seizure events in most cases.

However, as shown in Fig. 8, the performance of the seizure detection model varied across patients. For patients with low PPV but high

Table 5
Performance comparison between the proposed multimodal method and single-modality (ECG- or PPG-based) seizure detection models.

Research	Device	Classifier	Patients	Monitoring duration	Seizures	SEN	FAR /hour
Proposed method	ACM + PPG wearable armband	CNN+ LSTM with attention	68	783 nights (>8000 hours)	1846	0.716	0.165
Cooman et al. [24]	single-lead ECG	SVM+ transfer learning	24	2172 hours	227	0.710	1.900
Vandecasteele et al. [22]	clinical ECG device	SVM	11	701 hours	47	0.570	1.920
	wearable ECG device	SVM	11	701 hours	47	0.700	2.110
	wearable PPG device	SVM	11	701 hours	47	0.320	1.800

Note: SEN=sensitivity; FAR=false alarm rate; ACM=accelerometry; PPG=photoplethysmography; ECG=electrocardiography.

SEN, we found that they experienced more minor seizures compared to other patients. Our algorithm successfully detected these minor seizures but did not label them as severe seizure events, resulting in higher FAR. This discrepancy indicated that although our model was good at detecting seizure events, it sometimes overestimated their severity, resulting in more false positives. For the patient with a PPV of 0.0 and SEN of 0.0, reason for this result is that he/she had a very small number of seizures, only one or two in most cases. This limited number of events greatly affected the statistical reliability of performance metrics for this case, suggesting that the model's performance metrics might be less reliable for patients with extremely low seizure frequencies.

Our proposed method achieved a higher SEN while maintaining an acceptable FAR compared to existing studies. Most studies had faced challenges in balancing SEN and specificity, especially in reducing FAR without severely compromising the detection of true seizure events. Although the system exhibits relatively lower PPV, it was intentionally optimized for high SEN to minimize the risk of missing nocturnal seizures, given their potentially severe clinical consequences. In practical use, the detection threshold can be adapted to patient-specific needs and caregiver tolerance for false alarms, allowing customization of the SEN–PPV trade-off.

As shown in Table 5 , in the field of seizure detection, single ECG and HR-based sensors such as PPG had lower SEN and higher FAR. Specifically, when relying on only one wearable PPG sensor device, the SEN dropped to 0.320. This drop was attributed to the instability of the PPG signal during a seizure, where movements such as clonus could introduce instability and motion artifacts [31]. As a result, the PPG signal contained a lot of noise, making it impractical to extract reliable HR information by digital filtering. This result suggested that adding body movements as a supplement to the seizure detection process can significantly reduce FAR. However, this table presents a structured comparison of representative studies using different sensing modalities for seizure detection. As these studies vary in data sources and evaluation standards, the comparison is qualitative rather than quantitative. We fully recognize that comparing performance across studies is inherently difficult due to differences in patient populations, sensor modalities, recording environments, and annotation protocols. In particular, most existing studies rely on hospital-grade EEG data or short-term recordings under controlled conditions. These datasets often exclude ambiguous or hard-to-annotate cases, resulting in relatively balanced class distributions and potentially optimistic performance estimates.

In contrast, our study is based on a long-term, real-world wearable dataset collected in natural home environments, where seizure prevalence is extremely low and motion artifacts are common. Such conditions pose significant challenges for both detection performance and clinical applicability. To the best of our knowledge, there is currently no publicly available open-access dataset of long-term wearable recordings for seizure detection. This makes it impossible to perform standardized, cross-study model evaluation. Moreover, the algorithms developed for EEG-based detection may not generalize to wearable signals like PPG and ACM. Given these limitations, we aim to benchmark our framework specifically within the wearable sensor context, and we hope that this work can help promote the creation of open, continuous, and realistic seizure datasets collected from wearable devices. If regulatory conditions allow, such datasets would be crucial for enabling fair comparisons and accelerating advances in the field.

In the seizure detection step, we combined HR information (from PPG) and movement information (from ACM). By including PPG and HR signals, we not only proposed a novel DL-based seizure detection framework but also revealed a previously underexplored physiological insight: the onset of ictal HR changes typically precedes seizure-related motor activity during severe seizures. This temporal relationship, observed consistently across real-world multimodal recordings, offers valuable information for early and accurate seizure detection. To leverage this phenomenon, we incorporated a hybrid CNN-LSTM architecture with an attention mechanism, allowing the model to dynamically focus on temporal features across both PPG and ACM channels. This design enables the system to prioritize early cardiac abnormalities while capturing subsequent movement patterns, thereby improving detection sensitivity and reducing latency. However, when the PPG signal was of poor quality, the attention mechanism in the proposed detection model was expected to shift focus away from the PPG signal. In such cases, the model reduced the weight of the HR related channels when the ACM signal changed drastically. This adjustment resulted in a more stable detection effect. Therefore, the method proposed in this paper has a higher SEN and lower FAR in seizure detection. Our approach proved effective in facing the challenges posed by motion artifacts and noise, enhancing the overall reliability of seizure detection.

In addition to the main evaluation, we conducted a simplified single-modality analysis to assess the contribution of each input channel. Using only ACM could capture most convulsive seizures but missed cases where ictal HR changes occurred prior to motor activity. Conversely, using only PPG was highly vulnerable to motion artifacts, leading to an increased number of false alarms. These limitations were mitigated by integrating both modalities, enabling the framework to leverage complementary physiological information. This observation provides supportive evidence for the rationale of multimodal integration. These findings confirm that both the two-step strategy and the use of multimodal signals are essential in achieving a balance between accuracy, efficiency, and robustness in real-world wearable seizure detection. This work demonstrated a feasible multimodal, two-stage framework for nocturnal seizure detection in real-world wearable recordings. The development of more advanced fusion algorithms and a systematic quantification of multimodal advantages will be the focus of our future investigations.

Unlike many high-performing EEG-based seizure detection models evaluated on open-access datasets, our dataset consists of long-term, real-world recordings that are naturally imbalanced and noisy. Public datasets often contain carefully selected, well-annotated segments with relatively balanced seizure prevalence, which may not reflect the challenges of continuous monitoring. In contrast, real-world data may include only a few seizure events over several weeks, with significant motion artifacts and ambiguous signal patterns. These factors inherently limit model performance and increase the difficulty of achieving high PPV. Motion artifacts, transient signal dropouts, and PPG unreliability during movements could potentially affect the seizure detection. These effects were partially mitigated through preprocessing and the prescreening stage that filters out non-informative segments.

Furthermore, the attention mechanism embedded in the DL model helps dynamically focus on more informative temporal and channel features, thereby reducing the negative impact of noisy or unreliable PPG signals. Future deployments will integrate adaptive noise suppression techniques and redundant sensing modalities to enhance robustness under real-world conditions.

Despite these challenges, our method maintains acceptable sensitivity (SEN $= 0.716$) and a low false alarm rate (FAR $= 0.165$/h), which are clinically relevant for nocturnal seizure detection. In practice, while some false alarms are tolerable, especially for severe seizure detection, it remains important to continue optimizing the PPV to reduce caregiver burden. Therefore, we prioritize improving the FAR while retaining high sensitivity.

In our experimental evaluation, a 5-minute analysis window was employed to ensure sufficient feature extraction from ACM and PPG signals. This window was implemented as a historical (non-centered) window, meaning that seizure-related features at the beginning of the segment could trigger early detection without waiting for the full duration to pass. To address potential concerns about detection latency, we note that although a 5-minute analysis window was used in this study, the intended deployment is a real-time sliding-window system with partial overlap, which is initiated only after the pre-screening stage flags a suspected seizure segment. Detection is then triggered as soon as seizure activity enters the window, allowing for updates every few seconds and supporting timely nocturnal alerts in practical use. This means that the alarm does not necessarily require the entire 5-minute segment to elapse but can occur earlier depending on when the seizure activity enters the analysis window. Moreover, clinical studies have reported that caregiver response times to nocturnal seizures at home often range from 30 s to several minutes [32], suggesting that our framework's latency is within a clinically acceptable range.

Furthermore, our proposed framework is tailored for real-world deployment in wearable devices. By utilizing both PPG and tri-axial ACM signals, which were commonly available in commercial armbands, we ensure broad applicability without requiring EEG or complex clinical infrastructure. The two-step structure, including a lightweight pre-screening module and a deep classification model, reduces computational load by over 80 %, facilitating continuous long-term monitoring at home. These findings and architectural innovations demonstrate not only the scientific novelty but also the practical utility of our work in supporting independent living and proactive care for people with epilepsy.

However, there were several limitations of this study. First, the dataset, while wide-ranging, was derived from specific patients and they had highly imbalanced number of seizures. This might limit the generalizability of our findings. Additionally, the variability in seizure types among different patients suggested that a one-fits-all model might not be optimal. Future work will incorporate more diverse patient demographics and seizure phenotypes to improve the generalizability of the proposed approach. Our future work would also explore personalized models that are able to adapt to the characteristics of individual patients. It is also important to address data imbalances to improve the reliability and accuracy of models across all patients. Future studies should also focus on improving model specificity and reducing FAR without compromising SEN. Furthermore, data augmentation represents a promising direction for enhancing the robustness of seizure detection models. However, unlike in image domains, augmentation for physiological signals such as PPG and ACM requires careful consideration to preserve clinical and temporal validity. More advanced techniques, such as using generative models to simulate realistic seizure-like events, also require thorough validation. Given these complexities, we consider data augmentation a non-trivial yet important research direction and plan to address it in future dedicated studies.

In summary, our study demonstrates the feasibility of a two-stage multimodal seizure detection framework that balances computational efficiency and accuracy. By combining ACM and PPG/HR signals within an attention-enhanced CNN–LSTM model, the system captures complementary physiological cues and the temporal precedence of ictal HR changes, which enhances detection reliability and is an important underexplored physiological phenomenon. The use of a wearable armband further underscores the clinical value of enabling continuous, non-invasive nocturnal monitoring at home. This expands accessibility to a broader patient population, reduces caregiver burden, and addresses the urgent need for SUDEP risk mitigation during unsupervised night-time periods. Nevertheless, prospective validation in larger and more diverse patient cohorts will be essential to fully establish clinical effectiveness.

Beyond detection accuracy, the proposed method has the potential to contribute to understanding lifestyle-related seizure triggers. Commonly reported factors such as sleep deprivation, physical or emotional stress, hormonal changes, and alcohol or drug use are known to influence seizure onset in many patients. By enabling continuous home-based monitoring, our system could facilitate the identification of temporal patterns between lifestyle factors and seizure events. Integrating wearable data with self-reported behavioral logs in future studies may offer deeper insight into modifiable seizure risks and inform targeted intervention strategies aimed at seizure prevention.

6. Conclusion

In conclusion, our study presented a practical and effective approach for the long-term detection of severe nocturnal epileptic seizures using a wearable system that combines motion and cardiac signals. The proposed two-step method, incorporating both pre-screening and DL, delivers improved sensitivity and reduced false alarms compared to existing approaches. Despite certain limitations, the results indicated significant potential for improving patient care and warrant further investigation and refinement. Importantly, it provides new opportunities to investigate lifestyle-related seizure triggers in real-world settings, thereby advancing early detection and prevention strategies in epilepsy management. This work provides a promising approach toward scalable, accurate, and efficient seizure detection systems, advancing the integration of DL methodologies into wearable health monitoring applications.

Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

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Ethics approval

This study has the formal ethical approval from the Ethics Committee of Kempenhaeghe Epilepsy Center in the Netherlands, all participants or their legal guardians provided written informed consent.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

CRediT authorship contribution statement

Chunjiao Dong: Writing – original draft, Visualization, Software, Methodology, Funding acquisition. **Johannes P. van Dijk:** Supervision, Data curation, Conceptualization. **Ronald M. Aarts:** Writing – review & editing, Supervision, Conceptualization. **Yunfeng Wang:** Supervision, Investigation. **Xi Long:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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